

Title:

Oriented object detection in optical remote sensing images using deep learning: a survey

Authors:

Kun Wang, Zi Wang, Zhang Li, Ang Su, Xichao Teng, Erting Pan, Minhao Liu, Qifeng Yu

Summary:

This survey paper provides a comprehensive review of recent deep learning advancements in **oriented object detection** specifically for **optical remote sensing images**.

The paper outlines the evolution from horizontal to oriented object detection, detailing key challenges such as feature misalignment, spatial misalignment, oriented bounding box (OBB) regression problems, and general issues in remote sensing. It then categorizes and discusses existing deep learning methods based on detection frameworks, OBB regression techniques, and feature representation approaches, explaining how they tackle these challenges.

Additionally, the survey covers publicly available datasets, evaluation protocols, offers a comprehensive comparison of state-of-the-art methods, and identifies future research directions in the field.

Paper:

<http://arxiv.org/abs/2302.10473v6>

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Title:

Learn to Accumulate Evidence from All Training Samples: Theory and Practice

Authors:

Deep Pandey, Qi Yu

Summary:

This research paper addresses a significant limitation in Evidential Deep Learning (EDL) models, which are designed to make neural networks uncertainty-aware by quantifying fine-grained uncertainty using learned evidence.

While EDL offers a principled way to achieve this, it requires special activation functions to ensure the evidence is non-negative, a constraint that often leads to ****inferior predictive performance**** compared to standard softmax models, especially on large datasets.

The authors theoretically investigate this issue and identify a fundamental cause: ****existing evidential activation functions create "zero evidence regions," effectively preventing the model from learning from training samples that fall into these regions.****

Based on this theoretical understanding, they propose a ****novel regularizer**** designed to alleviate this limitation. Extensive experiments confirm their theoretical findings and demonstrate the effectiveness of their proposed approach in improving the performance of evidential models.

Paper:

<http://arxiv.org/abs/2306.11113v2>

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Title:

The Modern Mathematics of Deep Learning

Authors:

Julius Berner, Philipp Grohs, Gitta Kutyniok, Philipp Petersen

Summary:

This paper introduces the new field of mathematical analysis of deep learning, which aims to address research questions that classical learning theory could not explain. These questions concern the outstanding generalization power of overparametrized neural networks, the role of depth, the apparent absence of the curse of dimensionality, successful optimization despite non-convexity, the nature of learned features, the exceptional performance in physical problems, and the impact of architectural details on learning tasks. The paper provides an overview of modern approaches that offer partial answers to these challenges, detailing the main ideas of selected methods.

Paper:

<http://arxiv.org/abs/2105.04026v2>

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Title:

How reproducible are data-driven subtypes of Alzheimer's disease atrophy?

Authors:

Emma Prevot, Cameron Shand, Neil Oxtoby, for Alzheimer's Disease Neuroimaging Initiative

Summary:

This research paper investigates the reproducibility and robustness of data-driven Alzheimer's disease (AD) atrophy subtypes across different datasets.

****Objective:**** To evaluate the consistency of AD progression subtypes identified by the Subtype and Stage Inference (SuStain) algorithm, particularly how robust these definitions are when validated across diverse patient cohorts and when cognitively normal controls are included or excluded from the analysis.

****Methodology:**** The study utilized T1-weighted MRI data from 5,444 subjects drawn from three major datasets (ANMerge, OASIS, and ADNI), forming four independent cohorts. Each cohort was analyzed under two conditions: one including cognitively normal controls and one excluding them.

****Key Findings:****

- * The study confirmed the identification of three primary AD atrophy subtypes previously reported: Typical, Cortical, and Subcortical.
- * It also identified the emergence of rarer and atypical AD variants, such as posterior cortical atrophy (PCA).
- * Notably, the robustness of each subtype varied regarding the inclusion of controls, with certain subtypes (e.g., Subcortical) being more influenced by the overall cohort composition.

****Conclusion:**** The findings affirm SuStaln's reliability in defining stable AD subtypes, suggesting its potential utility in clinical stratification for trials and diagnosis. However, the study also highlights a critical need for improved dataset diversity, especially in terms of ethnic representation, to enhance the generalizability and broader clinical applicability of these subtypes.

Paper:

<http://arxiv.org/abs/2412.00160v1>

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Title:

A comprehensive review on Plant Leaf Disease detection using Deep learning

Authors:

Sumaya Mustofa, Md Mehedi Hasan Munna, Yousuf Rayhan Emon, Golam Rabbany, Md Taimur Ahad

Summary:

This paper presents a systematic review focusing on the detection of plant leaf diseases using deep learning models. It highlights that early diagnosis is crucial for managing these common and often fatal plant diseases. The review details the advantages and limitations of various deep learning architectures, including Vision Transformers (ViT), Deep Convolutional Neural Networks (DCNN), standard CNNs, Residual Skip Network-based Super-Resolution for Leaf Disease Detection (RSNSR-LDD), Disease Detection Network (DDN), and YOLO. Additionally, the paper examines how existing studies apply these models to publicly available datasets and utilize different performance metrics like accuracy, precision, and recall for evaluation.

Paper:

<http://arxiv.org/abs/2308.14087v1>

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Research Gap Analysis

Based on the provided summaries, here are the research gaps, limitations of current approaches, and directions for future work across these diverse deep learning domains:

1. Research Gaps

These are areas where understanding or methods are still largely undeveloped or incomplete.

1. ****Fundamental Mathematical Understanding of Deep Learning Phenomena:**** Despite practical success, the underlying mathematical principles explaining the outstanding generalization power of overparametrized networks, the role of depth, the absence of the curse of dimensionality, successful non-convex optimization, and the nature of learned features are still largely incomplete. (Paper: *The

Modern Mathematics of Deep Learning*)

2. ****Robust Uncertainty Quantification without Performance Degradation:**** While methods like Evidential Deep Learning (EDL) aim to quantify uncertainty, a fundamental cause for their inferior predictive performance compared to standard models on large datasets (due to "zero evidence regions") indicates a significant gap in developing uncertainty-aware models that can learn effectively from all data without compromise. (Paper: *Learn to Accumulate Evidence from All Training Samples: Theory and Practice*)
3. ****Generalizable and Ethically Representative AI Models:**** There's a critical gap in developing AI models (e.g., for medical diagnosis) that are robustly generalizable across diverse populations, especially concerning ethnic representation. Current models trained on limited datasets may not perform reliably on underrepresented groups. (Paper: *How reproducible are data-driven subtypes of Alzheimer's disease atrophy?*)
4. ****Integrated Multi-Modal Remote Sensing Object Detection:**** While oriented object detection in *optical* remote sensing is surveyed, there's an implicit gap in the comprehensive integration and survey of oriented object detection across other remote sensing modalities (e.g., Synthetic Aperture Radar (SAR), hyperspectral, multispectral images) which present different data characteristics and challenges. (Paper: *Oriented object detection in optical remote sensing images using deep learning: a survey*)
5. ****Real-World Robustness and Generalization in Applied Domains:**** For applied tasks like plant disease detection, there's a gap in developing models that can reliably perform in highly variable real-world conditions (e.g., varying lighting, occlusions, different plant varieties, early disease stages) beyond controlled dataset environments. (Paper: *A comprehensive review on Plant Leaf Disease detection using Deep learning*)

2. Limitations

These are specific shortcomings or constraints identified in existing methods or data.

1. ****Inferior Predictive Performance of Uncertainty-Aware Models:**** Existing Evidential Deep Learning (EDL) models often exhibit inferior predictive performance compared to standard softmax models, particularly on large datasets, primarily because current evidential activation functions

create "zero evidence regions" preventing the model from learning from certain training samples. (Paper: **Learn to Accumulate Evidence from All Training Samples: Theory and Practice**)

2. ****Dataset Diversity and Generalizability Issues in Medical AI:**** A critical limitation in medical imaging studies (e.g., Alzheimer's disease subtyping) is the lack of sufficient dataset diversity, especially in terms of ethnic representation. This hinders the generalizability and broader clinical applicability of identified disease subtypes and models. (Paper: **How reproducible are data-driven subtypes of Alzheimer's disease atrophy?**)

3. ****Specific Technical Challenges in Oriented Object Detection:**** Current deep learning methods for oriented object detection in remote sensing face several limitations, including:

- * ****Feature Misalignment:**** Difficulty in aligning features with arbitrarily oriented objects.
- * ****Spatial Misalignment:**** Issues with precisely localizing and extracting features for rotated objects.
- * ****Oriented Bounding Box (OBB) Regression Problems:**** Difficulties in accurately regressing the multiple parameters required for OBBs, often leading to performance bottlenecks.

(Paper: **Oriented object detection in optical remote sensing images using deep learning: a survey**)

4. ****Incompleteness of Theoretical Explanations:**** Classical learning theory is insufficient to explain many core phenomena observed in modern deep learning, such as the efficacy of overparameterization or the success of non-convex optimization. While new mathematical analyses offer "partial answers," a complete theoretical framework is still lacking. (Paper: **The Modern Mathematics of Deep Learning**)

5. ****Varying Robustness of Data-Driven Subtypes:**** The robustness of certain data-driven subtypes (e.g., AD atrophy subtypes) can be influenced by the overall cohort composition, such as the inclusion or exclusion of cognitively normal controls, suggesting a sensitivity to input data characteristics. (Paper: **How reproducible are data-driven subtypes of Alzheimer's disease atrophy?**)

6. ****Limitations of Specific Deep Learning Architectures in Applied Tasks:**** Various deep learning architectures (CNNs, ViTs, YOLO, etc.) applied to tasks like plant leaf disease detection have their own inherent advantages and limitations, implying that none are universally optimal and often struggle with specific challenges (e.g., fine-grained distinctions, robustness to real-world noise, computational cost) not fully resolved by current methods. (Paper: **A comprehensive review on Plant Leaf Disease detection using Deep learning**)

3. Future Work

These are explicit or strongly implied recommendations for future research and development.

1. ****Developing Novel Regularizers and Activation Functions for Uncertainty Quantification:**** Future work should focus on designing new regularizers or activation functions that alleviate "zero evidence regions" and allow evidential models to learn from all training samples, thereby improving predictive performance without sacrificing uncertainty quantification. (Paper: *Learn to Accumulate Evidence from All Training Samples: Theory and Practice*)
2. ****Enhancing Dataset Diversity for Medical AI:**** A critical future direction is to improve dataset diversity, particularly ethnic representation, in medical imaging datasets to ensure that data-driven subtypes and AI models are more generalizable and clinically applicable across broader populations. (Paper: *How reproducible are data-driven subtypes of Alzheimer's disease atrophy?*)
3. ****Advanced Methodologies for Oriented Object Detection:**** Future research in remote sensing should concentrate on developing novel deep learning architectures and techniques to overcome the identified challenges of feature misalignment, spatial misalignment, and OBB regression problems. This includes exploring more robust feature representations and regression strategies. (Paper: *Oriented object detection in optical remote sensing images using deep learning: a survey*)
4. ****Bridging Mathematical Theory and Practical Deep Learning Design:**** Future research should aim to further develop the mathematical analysis of deep learning, transforming partial answers into comprehensive theories that can directly inform the design of more efficient, robust, and interpretable deep learning algorithms. (Paper: *The Modern Mathematics of Deep Learning*)
5. ****Exploring Hybrid and More Robust Architectures for Applied Deep Learning:**** For tasks like plant disease detection, future work should investigate hybrid models or novel architectures that combine the strengths of different deep learning techniques (e.g., CNNs, ViTs, YOLO) to achieve higher accuracy, robustness, and efficiency in real-world, dynamic environments. (Paper: *A comprehensive review on Plant Leaf Disease detection using Deep learning*)
6. ****Expanding Oriented Object Detection to Multi-Modal Remote Sensing:**** Future work could extend the focus of oriented object detection beyond optical imagery to other remote sensing modalities, addressing the unique challenges and opportunities presented by SAR, hyperspectral, and multispectral data. (Paper: *Oriented object detection in optical remote sensing images using

deep learning: a survey*)

7. ****Translating Data-Driven Subtypes into Clinical Action:**** Beyond identifying stable subtypes, future work needs to focus on how these data-driven classifications (e.g., AD subtypes) can be effectively translated into actionable clinical strategies for diagnosis, prognosis, and personalized treatment stratification. (Paper: *How reproducible are data-driven subtypes of Alzheimer's disease atrophy?*)